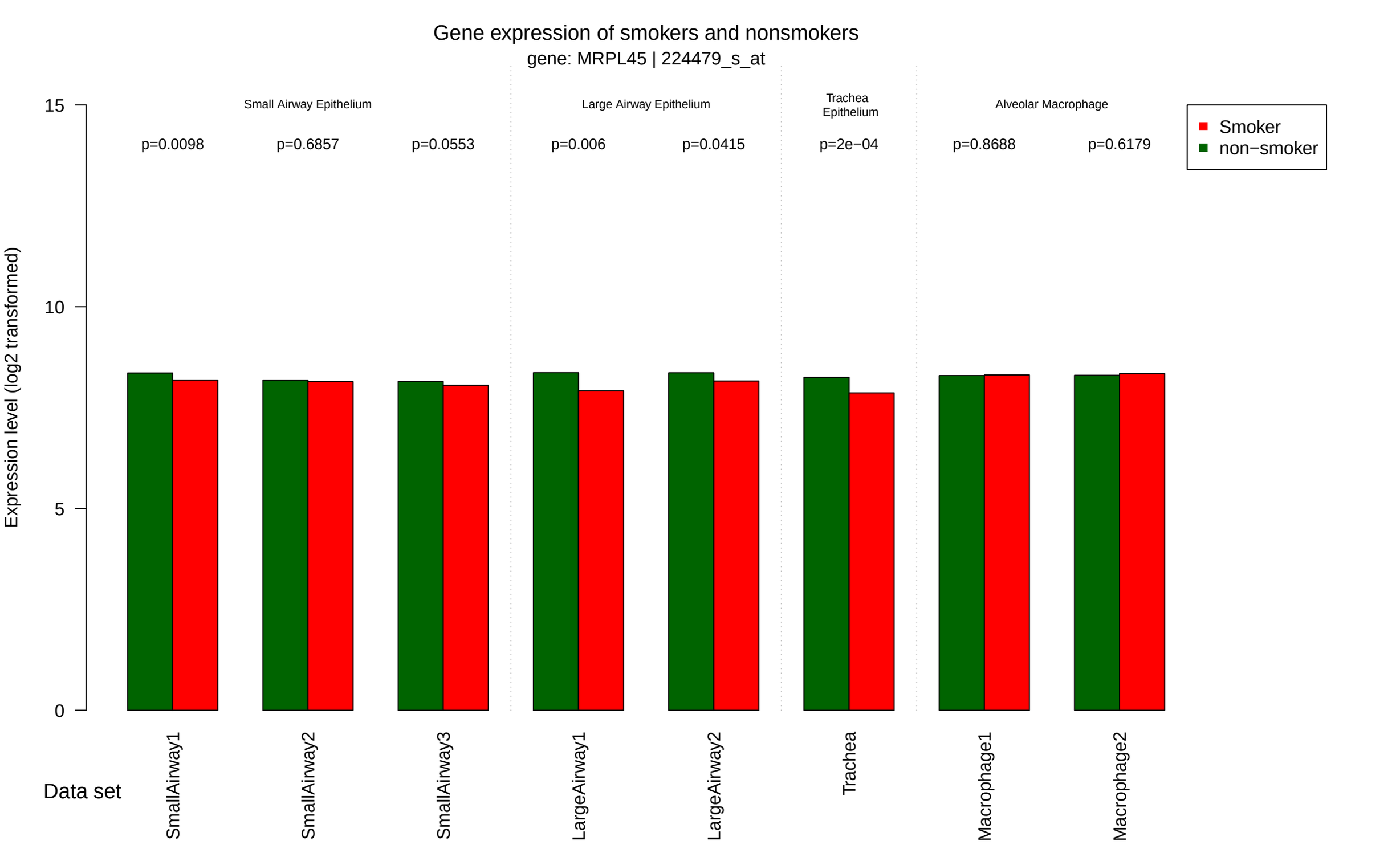
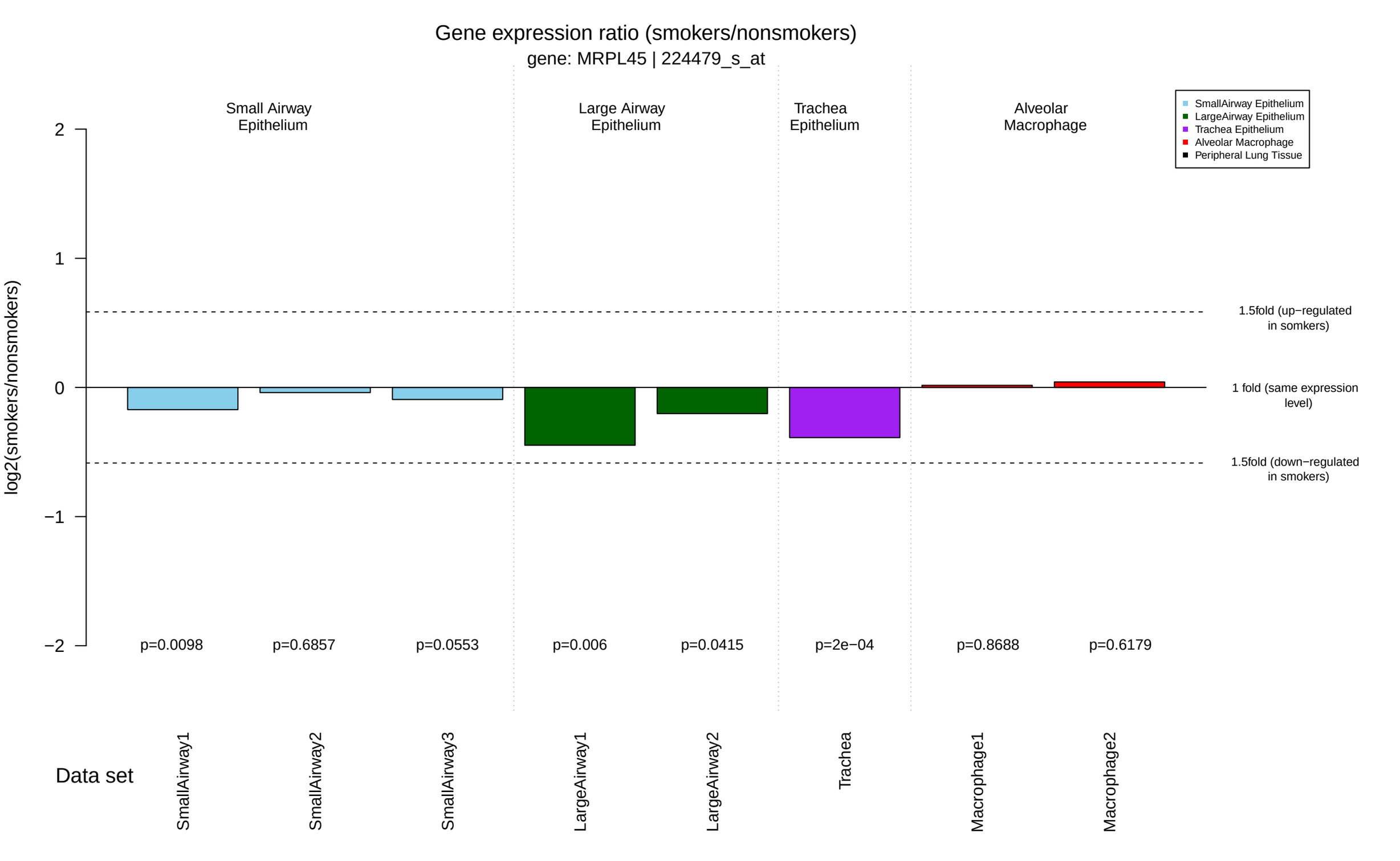
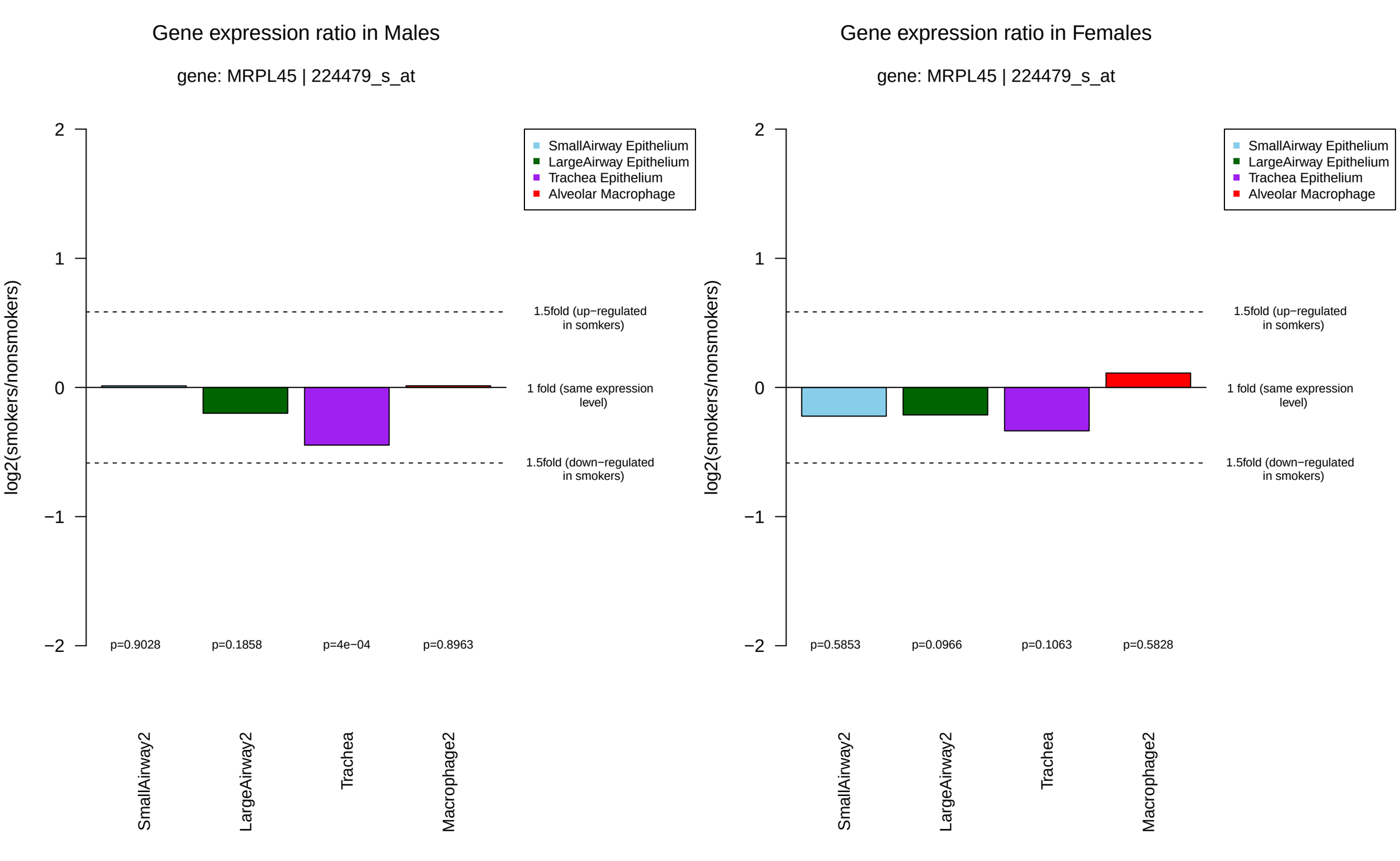
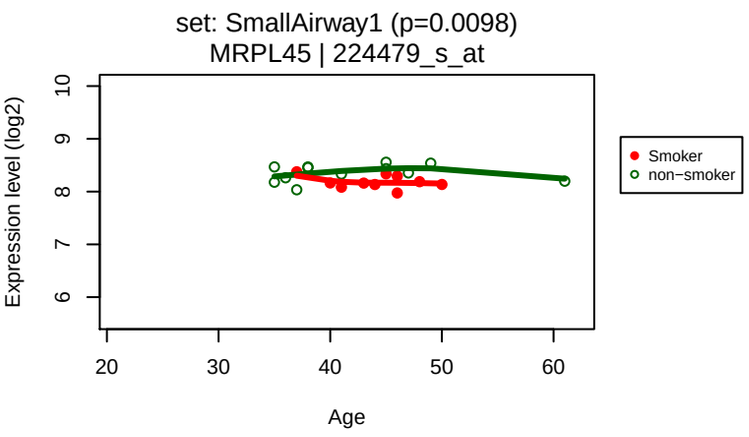
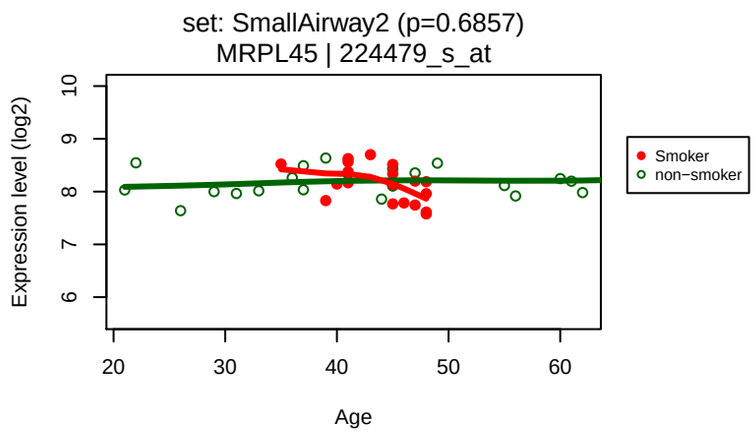
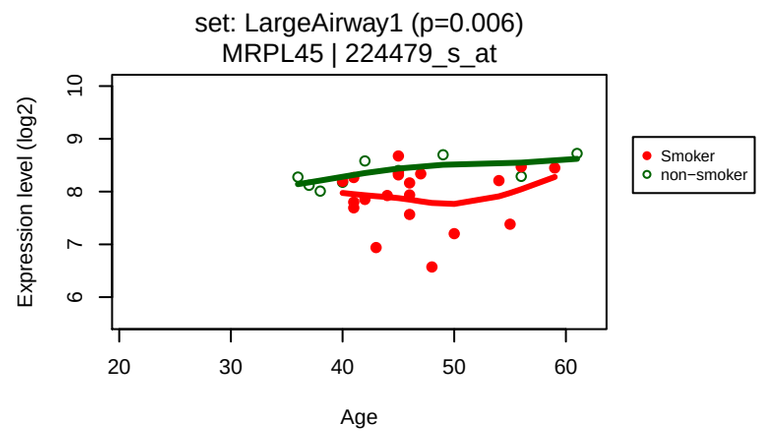

Gene expression of smokers and nonsmokers
gene: MRPL45 | 224479_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0098
p=0.6857
p=0.0553
p=0.006
p=0.0415
p=2e−04
p=0.8688
p=0.6179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

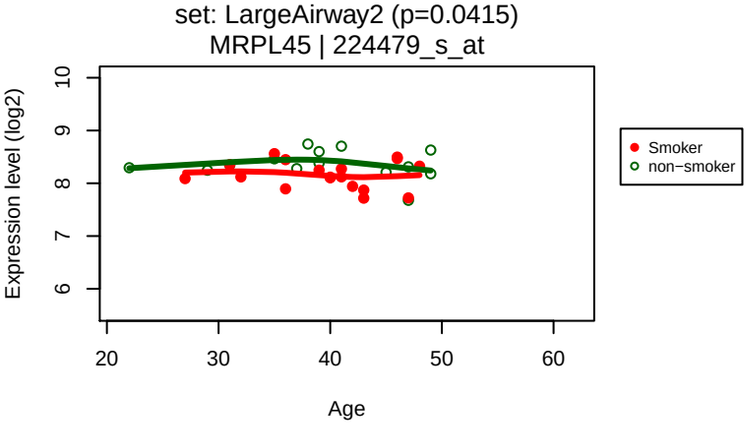

Gene expression ratio (smokers/nonsmokers)
gene: MRPL45 | 224479_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.6857
p=0.0553
p=0.006
p=0.0415
p=2e−04
p=0.8688
p=0.6179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

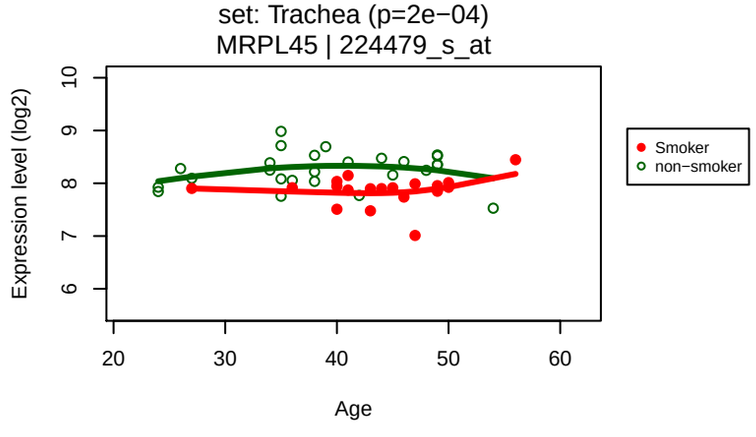

Gene expression ratio in Males
gene: MRPL45 | 224479_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9028
p=0.1858
p=4e−04
p=0.8963
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL45 | 224479_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5853
p=0.0966
p=0.1063
p=0.5828
SmallAirway2
LargeAirway2
Trachea
Macrophage2

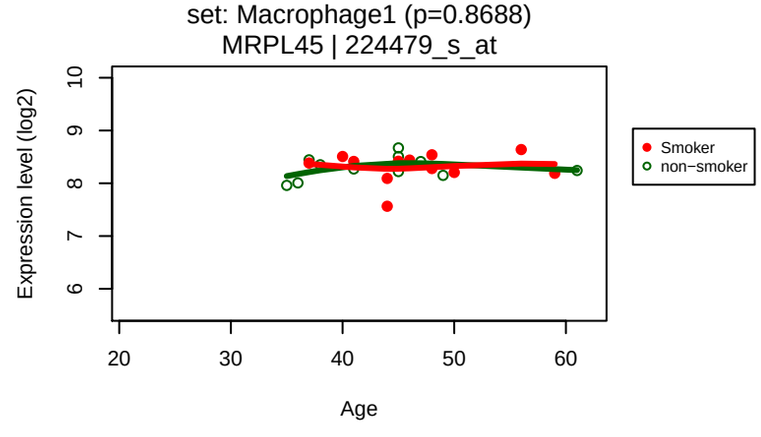

set: SmallAirway1 (p=0.0098)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker

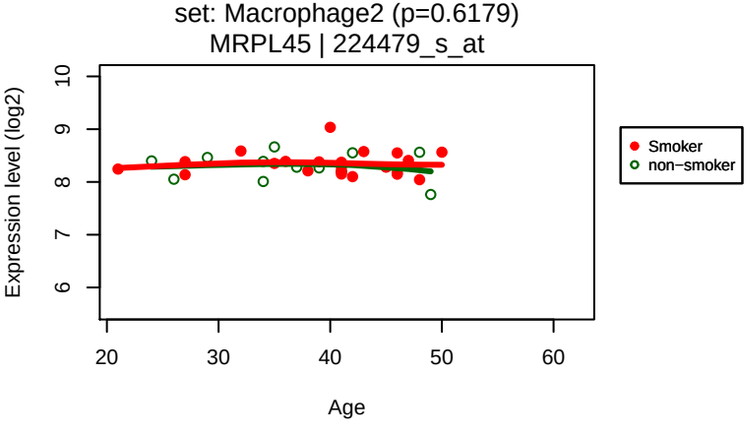

set: SmallAirway2 (p=0.6857)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker

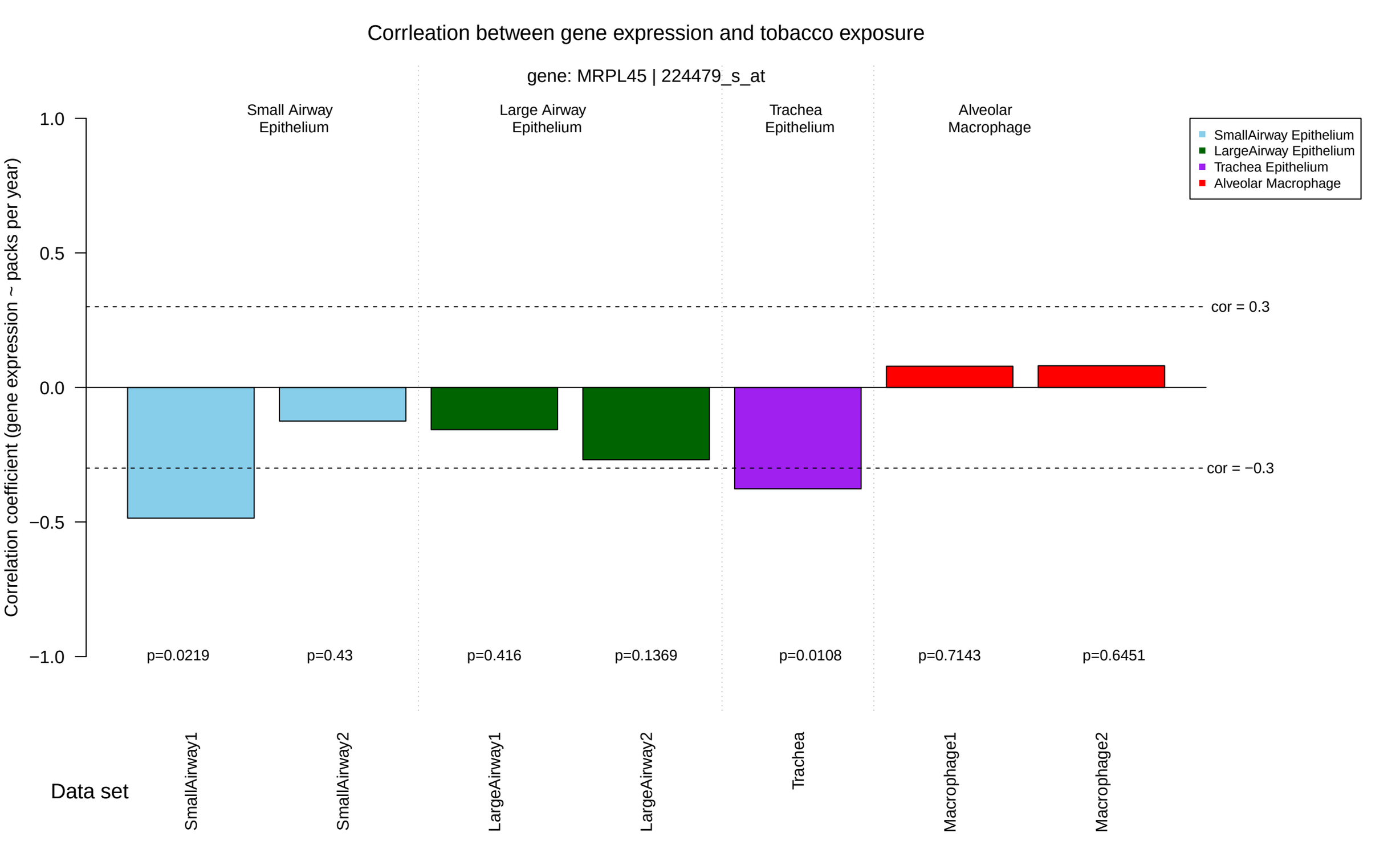

set: LargeAirway1 (p=0.006)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0415)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8688)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6179)
MRPL45 | 224479_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL45 | 224479_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0219
p=0.43
p=0.416
p=0.1369
p=0.0108
p=0.7143
p=0.6451
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2